Inputs

BOLD (COI) or UNITE (ITS)
reference barcodes, BINs or SHs

Open Tree
topology

TimeTree
divergence times

from published syntheses
from barcodes
related genera, for rooting only

age from TimeTree
age interpolated

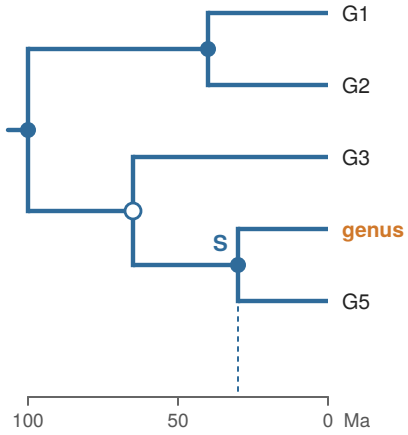
(i) Reference units

one representative sequence per BIN

filters: length, ambiguous bases, reading frame (COI)
genus = majority identification

(ii) Dated backbone

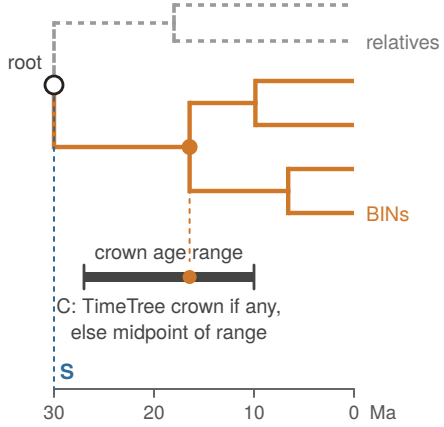
one tip per genus, ages in Ma



S: stem age of the genus

(iii) Dated subtree per genus

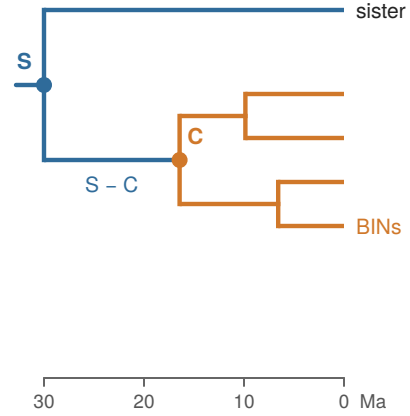
barcode tree, rooted on related genera



min: BIN divergence ÷ rate
max: 0.9 S and TimeTree crown

(iv) Grafting

genus tip replaced by its subtree



no rescaling: both parts in Ma;
genera run in parallel

Samples: placement and diversity

metabarcoding sequence (ASV) → best match among reference sequences

- **≥ 97%** assigned to that BIN
- **80–97%** placed on the tree by nearest BIN
- **< 80%** left out and reported

Diversity in Ma of history

Faith's PD, mean pairwise distance; between samples: UniFrac, PhyloSor